

# ELI — Installation

ELI v2.1.47

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## Installation — one-click setup, cross-platform (2026-07-05)

**Updated for v2.1.30 (July 2026).** The primary install path is now the prebuilt installers on GitHub Releases — the Windows Setup.exe and Linux AppImage are CI-launch-tested; the macOS dmg is built on a Mac, best-effort (not CI-verified) with first-boot GPU (CUDA/Vulkan/Metal) and starter-model offers; data lives in a per-user `ELI_v2` folder that survives upgrades. Source installs below remain fully supported.

One command per platform sets up Python deps, the **full SQLite architecture** (blank slate — schema only, no personal data), the **nomic embedder**, and the default voice.

| Platform         | Command                                                                                              | GPU                   |
|------------------|------------------------------------------------------------------------------------------------------|-----------------------|
| Linux            | <code>bash install.sh</code><br>( <code>--install-cuda</code> to also fetch the toolkit)             | CUDA                  |
| macOS            | <code>bash install.sh</code>                                                                         | Metal (no CUDA)       |
| Windows          | <code>install.bat / install.bat /cuda</code><br>( <code>→ install.ps1</code> )                       | CUDA (winget toolkit) |
| Android          | <code>bash</code><br><code>scripts/install_android.sh</code>                                         | CPU only (headless)   |
| Portable tarball | <code>./INSTALL_ELI.sh</code> then<br><code>./RUN_ELI.sh</code><br><code>--with-github-assets</code> | Same as host          |

## Which download should I get? (GitHub Releases)

Every release ships several downloads so you can trade **download size** against **how much works offline out of the box**. Two words explain them all:

- **lean** = source + launchers only (tiny). The Python dependencies **and** a model are fetched on first run, so the first launch needs internet once.
- **full** = the heavy bits are bundled so the first run is faster / more self-contained. On **Windows**, “full” bundles an **offline dependency wheelhouse** (no pip download). On **Linux**, “full” bundles a **starter model** (Qwen2.5-3B, ~1.9 GB) so no model download.

| Download                                              | Size    | Platform              | What’s inside                                 | First run needs internet for | Best for                                |
|-------------------------------------------------------|---------|-----------------------|-----------------------------------------------|------------------------------|-----------------------------------------|
| ELI-Setup-<v>.exe                                     | ~1 GB   | Windows               | Guided one-click installer                    | deps + model                 | Most Windows users                      |
| ELI_v2-<v>-windows-x64.zip                            | ~1.2 GB | Windows               | Prebuilt portable app (no install step)       | minimal                      | Windows, unzip-and-run                  |
| ELI_v2-<v>-windows-portable.zip ( <i>lean</i> )       | ~14 MB  | Windows               | Source + install.bat (online pip)             | deps + model                 | Windows, smallest download              |
| ELI_v2-<v>-windows-portable-full.zip ( <i>full</i> )  | ~950 MB | Windows               | Source + <b>offline dependency wheelhouse</b> | model only                   | Windows on locked-down/offline networks |
| ELI_v2-<v>-macos-arm64.dmg                            | ~1.2 GB | macOS (Apple Silicon) | App bundle                                    | model                        | Mac users                               |
| ELI_v2-<v>-linux-portable.tar.gz ( <i>lean</i> )      | ~7 MB   | Linux                 | Source + INSTALL_ELI.sh / RUN_ELI.sh          | deps + model                 | Linux, smallest download                |
| ELI_v2-<v>-linux-portable-full.tar.gz ( <i>full</i> ) | ~1.9 GB | Linux                 | Source + <b>bundled Qwen2.5-3B model</b>      | deps only                    | Linux, works with a model immediately   |
| ELI_v2-<v>-x86_64.AppImage ( <i>lean</i> )            | ~7 MB   | Linux                 | Double-click app; first launch self-installs  | deps + model                 | Linux, one-click, small download        |
| ELI_v2-<v>-x86_64-full.AppImage ( <i>full</i> )       | ~1.9 GB | Linux                 | Double-click app + <b>bundled model</b>       | deps only                    | Linux, one-click, model included        |
| eli_v2_0-<v>-py3-none-any.whl                         | ~2 MB   | any                   | pip wheel (pip install ...whl[full])          | deps + model                 | Developers / CI / existing venvs        |
| SHA256SUMS.txt                                        | tiny    | —                     | SHA-256 for every file above                  | —                            | verifying your download                 |

**In plain terms:** - Want it to just work and don’t mind a one-time first-run download → **Setup.exe** (Windows), **.dmg** (macOS), or the **AppImage (lean)** (Linux). - On a slow or locked-down network → the **-full** variant for your OS (fewer downloads later). - On **Linux** the *model* is the big download, so the

-**full** tar / AppImage bundle it (the Python deps still install via pip on first run — internet needed once). On **Windows** the *deps* are the awkward part, so **windows-portable-full** bundles those (the model still downloads). Neither “full” is 100 % offline; each bundles the piece that’s hardest to fetch. - Verify any download against the checksums: `sha256sum -c SHA256SUMS.txt` (Linux/macOS).

**Model size vs. your GPU matters more than which download you pick.** Choose a model that fits your VRAM — e.g. **Qwen3-8B** / **Qwen2.5-7B** on an 8 GB GPU. Very large models (30B+) run mostly on CPU and will be slow, and the local **web server** may appear to hang on the first message while such a model loads.

## Linux — Arch and other lean distros

The **AppImage** is the easiest path on any distro: it bundles its own **Python 3.11** (so Arch’s system Python 3.14, which has no `llama-cpp-python` wheel, is irrelevant) and, since **v2.1.21**, every Qt xcb library it needs — so it launches out of the box with no extra packages. Download and run it **directly**:

```
U=https://github.com/ShadowESC95/ELI_v2.0/releases/download/v2.1.30
wget "$U/ELI_v2-2.1.30-x86_64.AppImage"
chmod +x ELI_v2-2.1.30-x86_64.AppImage
./ELI_v2-2.1.30-x86_64.AppImage
```

Two fixes worth knowing, both resolved in current builds and verified on a clean Arch VM:

- **“attempt to write a readonly database” (portable build).** If ELI is extracted onto an **NTFS** / **exFAT** / **FAT** or **network** filesystem — e.g. a `~/Downloads` on a dual-boot data drive — SQLite’s WAL journal isn’t supported there and database init fails (only the `user.*` stores; the others in the same folder succeed, which is the tell-tale). **v2.1.19+** detects a WAL-hostile filesystem and falls back to a rollback journal automatically. On older builds, extract onto an **ext4/btrfs** path instead. Check a mount with `findmnt -T . -o TARGET,FSTYPE`.
- **GUI won’t open — “libxcb-cursor0 ... xcb platform plugin”.** Qt 6.5+ needs the whole **xcb-util family**, which a minimal desktop doesn’t ship (Qt’s error is misleading — the real missing library is often `libxcb-icccm.so.4`, not the cursor lib). **v2.1.21+** bundles all of them in the AppImage. On an older build or a **source** install, add them:
  - Arch: `sudo pacman -S xcb-util-cursor xcb-util-wm xcb-util-image xcb-util-keysyms xcb-util-renderutil xcb-util`
  - Debian/Ubuntu: `sudo apt install libxcb-cursor0 libxcb-icccm4 libxcb-image0 libxcb-keysyms1 libxcb-render-util0 libxcb-util1`

Run the AppImage **directly** (as above) rather than with `--appimage-extract-and-run`, which unpacks ~4 GB into `/tmp` and can fail with `libz.so.1: file too short` on a small tmpfs. Launching over **SSH** into a running desktop session? Prefix with `export DISPLAY=:0`. And paste long download URLs as a **single line** — a wrapped URL 404s and its tail runs as a stray command.

## Desktop / applications-menu icons

On its **first launch** the Linux AppImage offers to add **ELI**, **ELI Server** and **Uninstall** to your applications menu (with the app icon), pointing the launchers at the `.AppImage` file. This works when you run the **.AppImage directly** or via `--appimage-extract-and-run` — both set the `APPIMAGE` path the launcher needs. Force it any time with:

```
./ELI_v2-2.1.30-x86_64.AppImage --integrate      # add/refresh menu entries
./ELI_v2-2.1.30-x86_64.AppImage --uninstall    # remove them
```

Running the **manually extracted** `./squashfs-root/AppRun` does *not* create menu icons — there’s no `.AppImage` file for the launcher to point at, so it’s skipped (with a one-line note on `stderr` explaining why). Use the real `.AppImage` for menu integration. On Ubuntu 24.04 a normal double-click/FUSE run needs `libfuse2` (`sudo apt install libfuse2t64`). The **portable tarball** and **source** installs create the same menu entries via `scripts/install_desktop_apps.sh`; on **Windows** the `Setup.exe` adds Start-menu shortcuts, and on **macOS** the app is the `.app` bundle you drag to Applications.

## CUDA toolkit option (--install-cuda / /cuda)

For non-technical users with an NVIDIA GPU but no toolkit. Best-effort, never fatal: - **Linux:** tries no-sudo pip nvidia-cuda-nvcc-cu12 (exposes nvcc via CUDACXX), then the system package manager (apt/dnf/pacman, if sudo is available), then prints the manual step — then source-rebuilds llama-cpp with -DGGML\_CUDA=on. - **Windows:** winget install Nvidia.CUDA, then rebuilds llama-cpp. - **macOS/Android:** N/A (Metal / CPU). The default install already uses prebuilt CUDA wheels (no toolkit needed); the option only matters when those don't match the user's CUDA or a source build is required.

## What bash install.sh does

1. Detects Python (3.10+) and OS; creates .venv; upgrades pip/setuptools/wheel.
2. Installs **PyTorch** (CUDA 12.1 / CPU / macOS-MPS per flags/OS).
3. Installs **llama-cpp-python** with GPU acceleration (CUDA wheel index / Metal / CPU) — then **verifies llama\_supports\_gpu\_offload()** and, if it landed CPU-only, prints the exact CUDA-rebuild command (closes the silent-CPU-wheel trap).
4. Installs the ELI package ([full]) + all remaining dependencies from the **frozen requirements.lock.txt** (exact known-good versions; reproducible).
5. Seeds config/settings.json from the template (offline-by-default, wizard on) — never overwrites an existing config.
6. Runs **python -m eli.core.init\_data** — creates **every** SQLite store/table (user.sqlite3, system\_index.sqlite3, coding\_memory.sqlite3, agent.sqlite3) with **zero personal memories/profile/history** (blank slate). System inventory (installed apps, \$PATH binaries) is scanned once so “open Firefox” works — that is machine environment data, not user memories.
7. Verifies import eli, the GUI entry, and the eli console script.
8. Fetches **nomic-embed-text-v1.5.Q4\_K\_M.gguf** → models/embeddings/ (~80 MiB, required for memory/RAG) via python -m eli.core.model\_download --aux.
9. Fetches **voice weights:** Piper en\_US-amy-medium + faster-whisper small.en via python -m eli.runtime.voice\_assets (idempotent; skipped if already present).
10. Optionally downloads a **chat GGUF** (wizard / --model= / --auto).
11. Prints how to launch + how to add more models.

## Three local asset layers (not in git)

| Asset                         | Path                                                | Size (typical)                   | When fetched                         |
|-------------------------------|-----------------------------------------------------|----------------------------------|--------------------------------------|
| <b>Embedder</b> (required)    | models/embeddings/nomic-embed-text-v1.5.Q4_K_M.gguf | ~80 MiB                          | install.sh / wizard / --aux          |
| <b>Chat model</b> (pick one+) | models/*.gguf                                       | 0.6-8+ GiB                       | Wizard / model_download / asset pack |
| <b>Voice</b> (default amy)    | models/tts/piper/ + tts_piper/piper/                | ~60 MiB Piper + ~464 MiB whisper | voice_assets / asset pack            |

GitHub Release tag **local-assets-v2.1** mirrors embedder + starter chat GGUFs + cleared Piper voices. Restore: ./RUN\_ELI.sh --with-github-assets. Voices excluded from auto-restore: ryan, lessac, cori — see models/MODEL\_LICENSES.md.

## Flags

- --cpu-only — no CUDA (CPU torch + CPU llama-cpp).
- --latest — use requirements.txt version ranges instead of the frozen lock.
- --skip-torch — leave an existing torch in place.
- --no-model — skip chat-model offer; embedder still fetched unless you disable network.

## Files

- `install.sh` (Linux/macOS), `install.bat` / `install.ps1` (Windows).
- `requirements.lock.txt` — frozen exact versions (excludes torch/llama-cpp, which install via their CUDA indices).
- `pyproject.toml` — package metadata + `[project.scripts]` (`eli` → `eli.gui.app:main`).

## Launch

`./eli.sh` or `source .venv/bin/activate && eli`. First run shows the **setup wizard** if no chat GGUF is present. Wizard also verifies embedder + voice and can fetch them.

Chat model: `python -m eli.core.model_download --auto` (or `--list`, or a named model). ELI stays offline by default; downloads are deliberate one-time actions.